Introduction

For many years now, you have been solving *algebraic equations*, such as

$$2x + 1 = 0,\ x^2 + 3x = 1,\ e^x = 7$$

and so on. Today, we will begin to examine *differential equations*. First, let's think about what we already know:

What does it mean to *solve* an algebraic equation? For example, what does it mean to solve $x^2 + 3x + 2 = 0$?

How do you *check* a proposed solution to an algebraic equation? For example, how do you check if $x = -2$ is a solution to $x^2 + 4 = -x^3$?

Is it easier to *solve* an algebraic equation, or to check a proposed solution? Explain.

Is the point $x = 1$, $y = 3$ (i.e. the point $(1, 3)$) a solution to the equation $2y + x = y + 4$? Why?

Is the point $(2, 3)$ a solution to the equation $y^2 + x = y + 8$? Why?

Is the point $(1, 1)$ a solution to the equation $x^2 = 1 - y^2$? Why?

What does it mean to say that a point (a, b) is a solution to an equation?

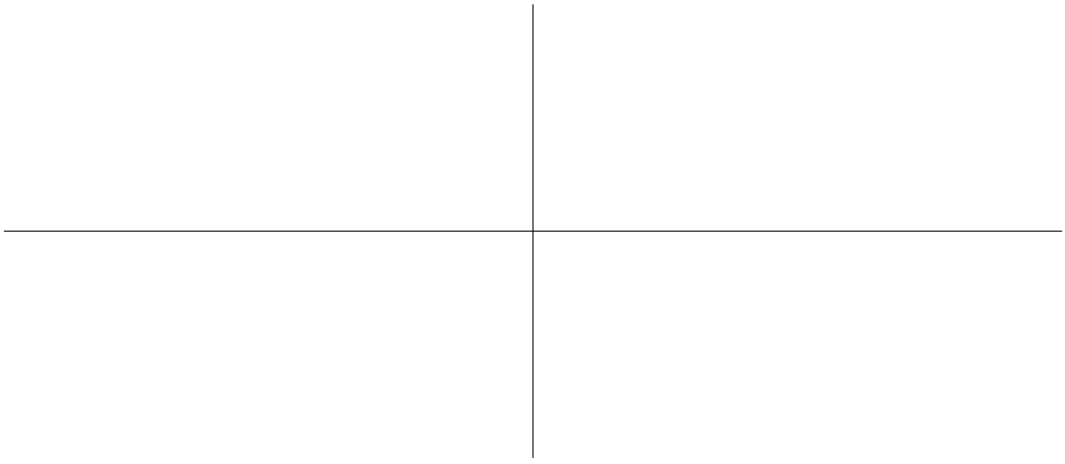
Differential Equations

Definition Equations of the form $\frac{dy}{dx} = y^2$, $\frac{dy}{dx} = 2x$ or $\frac{dy}{dx} = y + x$ are called ***differential equations***. In fact, any equation of the form $\frac{dy}{dt} = g(y, t)$ is a differential equation. $y = f(t)$ is a ***solution*** if when $f(t)$ is substituted for y in the expression $g(y, t)$, the result is $\frac{dy}{dt}$. In other words, like any other equation, when you substitute your answer into both sides of the equation, you get a true statement.

Differential Equations and Antidifferentiation

Check if the function $y = x^2$ a solution to the equation $\frac{dy}{dx} = 2x$. Why?

Consider the differential equation $\frac{dy}{dx} = \cos x$. Use antidifferentiation to find a solution of this equation. Can you find more than one solution? How many can you find? Graph a few of them on the set of axes below. How are they related to each other?



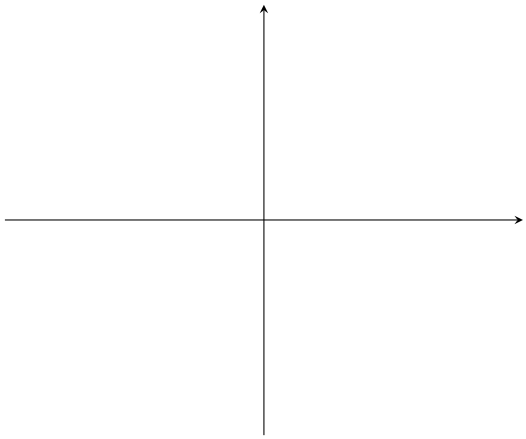
Definition The ***order*** of a differential equation is the highest derivative that appears in it. For example $\frac{dy}{dx} = x + y$ is a *first order* equation; $\frac{d^2y}{dx^2} + \frac{dy}{dx} = y$ is a *second order* equation.

1. Consider the second order differential equation $\frac{d^2y}{dx^2} = e^{-x}$.

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(a) Use antidifferentiation (twice) to find all the solutions of this equation. Part of your solution should be something like $Cx + D$. If its not, you did something wrong!

(b) Graph solutions on the axes below. Label each with the values of C and D you chose. You should see that the relationship between them is more complex than in the previous question. If you don't see that, choose different sets of values!



In general, differential equations do not have unique solutions. In fact, a differential equation often has an infinite number of solutions, as you saw above. The reason is that we antidifferentiate, we introduce an arbitrary constant: if $F(x)$ is an antiderivative of $f(x)$, then so is $F(x) + C$ for any value of C. When you are asked to *solve* a differential equation, you are required to find the *general solution*. That is, the solution with a constant in it.

Definition An *initial value problem* is an a differential equation with a specified value of the solution provided. Such a value is called an *initial condition*. Initial value problems most commonly have a unique solution.

2. (a) Solve the differential equation $\frac{dy}{dx} = 2e^x$.

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(b) Use your solution above to solve the initial value problem $\frac{dy}{dx} = 2e^x$, $y(0) = 3$.

3. (a) Solve the differential equation $\frac{d^2y}{dx^2} = 2\sin x$.

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(b) Use your solution above to solve the initial value problem $\frac{dy}{dx} = 2\sin x$, $y(0) = 3$, $y'(0) = 0$.

Above, we saw differential equations of the form $\frac{dy}{dx} = f(x)$. We found that solving such an equation is just antidifferentiating $f(x)$. That is, the general solution is

$$y(x) = \int f(x)\, dx.$$

If we had a second derivative involved, we needed to antidifferentiate twice to find a solution.

As we'll see in the coming days, solving differential equations is not always as straightforward as that, though!

More Problems

4. Solve the following initial value problems by antidifferentiation:

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(a) $\frac{dy}{dx} = x^2$, $y(1) = 1$.

(b) $\frac{d^2h}{dt^2} = -32$, $h(0) = 100$, $h'(0) = 10$. (See the Newton's Law of Motion lab!).

(c) $\frac{dy}{dx} = x\cos(x)$, $y(\pi) = 2$.

5. What is the order of the differential equation $\frac{d^3y}{dx^3} = 24x$? Find its the general solu-
□Regrade tion by antidifferentiating three times. Check that your solution has three arbitrary constants in it! Solve the initial value problems:

(a) $\frac{d^3y}{dx^3} = 24x$, $y(0) = 1$, $y'(0) = 2$, $y''(0) = 3$.

(b) $\frac{d^3y}{dx^3} = 24x$, $y(0) = 1$, $y(1) = 2$, $y(2) = 3$.